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Annex B

DS.02.02

Server Roles and Capabilities

Student Outline

Data Systems Administrator Course
M09CVQ1
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Note- During the labs in the class room you can play with removing domain controllers and the associated metadata clean up but in the CAPSTONE event the instructor in charge of your room will be the one to remove the metadata. If a domain controller does not install properly or you have to delete the Domain controllers VM notify the Instructor to do the cleanup. 35
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Terminal Learning Objective

1. Given a command's mission, threat assessment, operational environment, operational plans, virtual environment, and personnel, establish operating systems in accordance with applicable STIGs and local cybersecurity policies in support of the communications plan. (0671-IMPL-1002)

Enabling Learning Objectives

1. With the aid of reference and given the proper equipment and software, configure basic server configurations, in accordance with 0671-IMPL-1002.

2. With the aid of reference and given the proper equipment and software, connect a server to a network, in accordance with 0671-IMPL-1002.

3. With the aid of reference and given the proper equipment and software, test server connectivity, in accordance with 0671-IMPL-1002.

4. With the aid of reference and given the proper equipment and software, change a server's computer name, in accordance with 0671-IMPL-1002.

5. With the aid of reference and given the proper equipment and software, configure a server's firewall settings, in accordance with 0671-IMPL-1002.

6. With the aid of reference and given the proper equipment and software, deploy VMware Tools on a basic server, in accordance with 0671-IMPL-1002.

7. Without the aid of reference, identify in writing operating system server roles, in accordance with 0671-IMPL-1002.

8. With the aid of reference and given the proper equipment and software, deploy operating system server roles, in accordance with 0671-IMPL-1002.

9. Without the aid of reference, install Active Directory Domain Services in accordance with 0671-IMPL-1002.

10. Without the aid of reference, operate and maintain Active Directory Domain Services in accordance with 0671-IMPL-1002.

11. Without the aid of reference, install Active Directory Users and Computers in accordance with 0671-IMPL-1002.

12. Without the aid of reference, operate and maintain Active Directory Users and Computers in accordance with 0671-IMPL-1002.

13. Without the aid of reference, install Active Directory Sites and Services in accordance with 0671-IMPL-1002.

14. Without the aid of reference, operate and maintain Active Directory Sites and Services in accordance with 0671-IMPL-1002.

15. Without the aid of reference, install DNS in accordance with 0671-IMPL-1002.

16. Without the aid of reference, operate and maintain DNS in accordance with 0671-IMPL-1002.

17. Without the aid of reference, identify in writing operating system server features, in accordance with 0671-IMPL-1002.

18. With the aid of reference and given the proper equipment and software, deploy operating system server features, in accordance with 0671-IMPL-1002.

19. Without the aid of reference, install Remote Server Administration Tools in accordance with 0671-IMPL-1002.

20. Without the aid of reference, operate Remote Server Administration Tools in accordance with 0671-IMPL-1002

21. Without the aid of reference, create security groups in accordance with 0671-IMPL-1002

22. Without the aid of reference, maintain security groups in accordance with 0671-IMPL-1002

23. Without the aid of reference, demonstrate how to create objects in Active Directory Organizational Unit Structure in accordance with 0671-IMPL-1002

24. Without the aid of reference, demonstrate how to pre-stage objects in Active Directory Organizational Unit Structure in accordance with 0671-IMPL-1002

25. Without the aid of reference, maintain objects in Active Directory Organizational Unit Structure in accordance with 0671-IMPL-1002

To properly install a server role on an existing server, please refer to Annex B Lab 3 prior to implementation

Server Roles

A server role is a set of software programs that, when they are installed and properly configured, lets a server perform a specific function for multiple users or other computers within a network. Generally, roles share the following characteristics. They describe the primary function, purpose, or use of a server.

How many roles can a server have?

A server can have at most 250 roles. Depending on the amount on resources available, most servers may have up to three different roles. There are other times where each server will have one role installed. This is all planned out during the initial configurations and implementation.

Active Directory Domain Services

Active Directory Domain Services (AD DS) is a server role in Active Directory that allows admins to manage and store information about resources from a network, as well as application data, in a distributed database.

AD DS helps admins manage network elements -- both computing devices and users -- and reorder them into a custom hierarchical structure. AD DS also integrates security by authenticating logons and controlling access to directory resources. All types of USMC networks utilize AD DS in some fashion or form whether on tactical or in garrison.

How is Active Directory Domain Services used?

Active Directory is a directory service that runs on Microsoft Windows Server. It is used for identity and access management. AD DS stores and organizes information about the people, devices and services connected to a network. AD DS serves as a locator service for those objects and as a way for organizations to have a central point of administration for all activity on the organizations network.

AD DS is used in on-premises Windows environments, and Microsoft Azure AD DS is used in cloud-based Windows environments. They can be used together in hybrid cloud environments

How does AD DS work?

AD DS is the core component of Active Directory that enables users to authenticate and access resources on the network. Active Directory organizes objects into a hierarchy, which lets various Domain Services connect with them and users access or manage them. The hierarchical structure includes the following:

- **Domains.** A group of objects, such as users or groups of devices that share the same AD database makes up a domain.

- **Organizational units.** Within a domain, organizational units are used to organize objects within the domains.

- **Active Directory trees.** Multiple domains grouped together in a logical hierarchy make up an AD tree. The bonds between domains in a tree are known as "trusts."

- **Active Directory forests**. This AD functional level is made up of multiple trees grouped together. Trees in an AD forest share trusts, just like domains in a tree share trusts. Trusts enable constituent parts of a tree or forest to share things like directory schemas and configuration specifications.

What services does AD DS provide?

Active Directory covers a range of services. AD Domain Services is the main service that encompasses these five services.

- Domain Services

Domain Services stores centralized directory information and lets users and domains communicate. When a user attempts to connect to a device or resource on a network, this service provides login authentication, verifying the user's login credentials and access permissions.

- Lightweight Directory Services (LDS)

AD LDS is similar to Domain Services, but it uses Lightweight Directory Access Protocol (LDAP), which has fewer restrictions. AD LDS enables cross-platform capabilities that, for instance, let Linux-based computers function on the network.

- Active Directory Federation Services (AD FS)

AD FS provides single sign-on authentication, enabling users to sign in once to access multiple applications in the same session.

- Rights Management

This service controls data access policies and provides access rights management. For example, Rights Management determines which folders users can access.

- Certificate Services

Certificate Services allows the domain controller to create and manage digital certificates, signatures and public key cryptography.

What are the benefits of Active Directory Domain Services?

The four key benefits of AD DS include the following:

- Hierarchical structure. This is the main benefit of AD DS, providing the organizational structure for the information contained in Active Directory.

- Flexibility. AD DS gives users flexibility in determining how data is organized on the network. It simplifies administrative tasks by centralizing services like user and rights management and provides some security. Users can access Active Directory from any computer on the network.

- Single point of access. Domain Services creates a single point of access to network resources. This lets IT teams collaborate more efficiently and limit the access points to sensitive resources.

- Redundancy. AD DS has built in replication and redundancy. If one domain controller fails, another automatically takes over its responsibilities.

Installing AD DS

There are several scenarios where an administrator would install a Domain Controller (AD DS Role).

- First domain controller in a new forest
- Deploying domain controllers in an existing forest
- First domain controller in a new domain
- Additional domain controllers in a domain
- Read-only domain controllers

In the labs we will be focusing on the primary and secondary domain controllers.

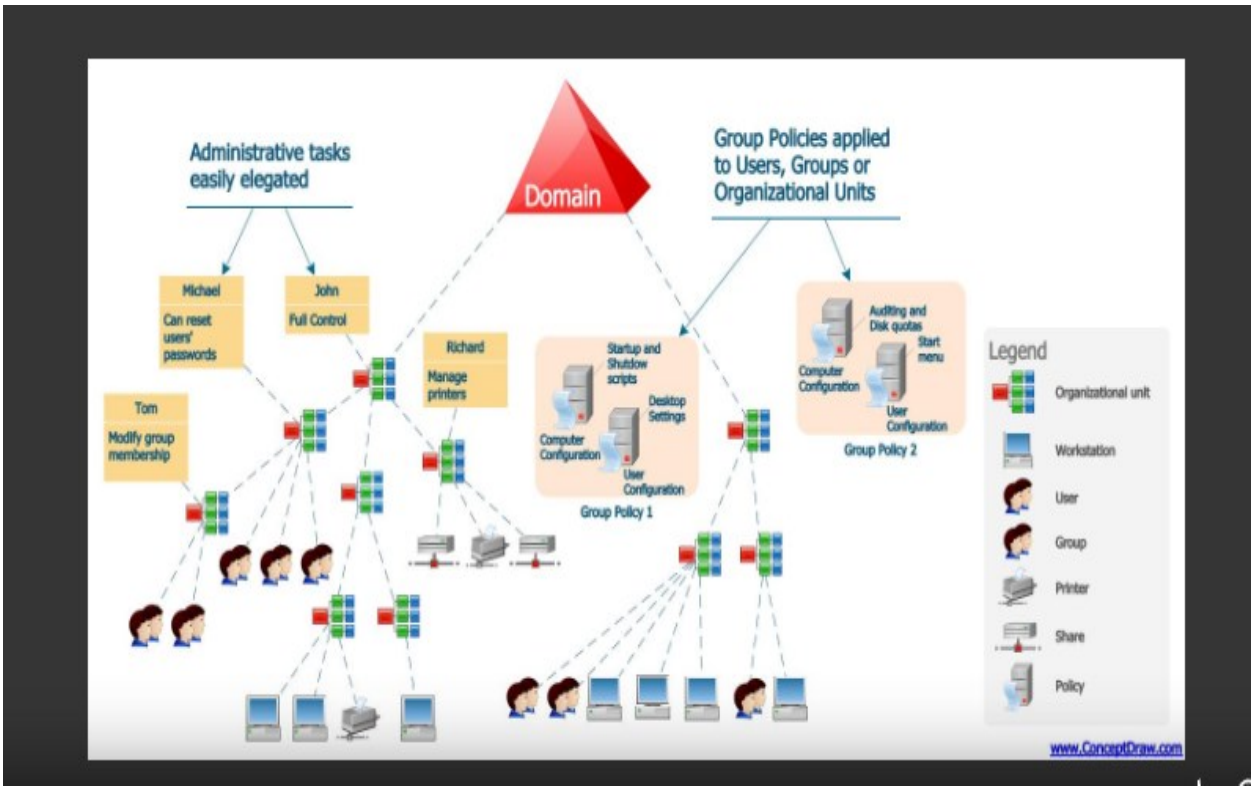
First Domain Controller (Primary)

Will not be joined to a domain because it is the domain.

Secondary Domain Controller

It is easier to join server to the domain before promoting it to a domain controller.

Note- In the Capstone Event you will only be creating secondary domain controllers.



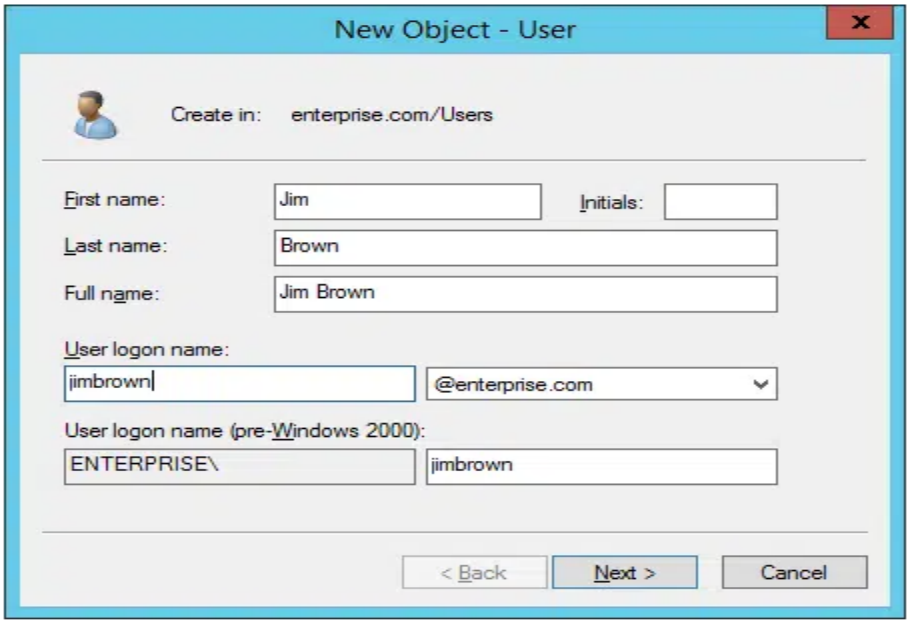
Active Directory Users and Computers (ADUC)

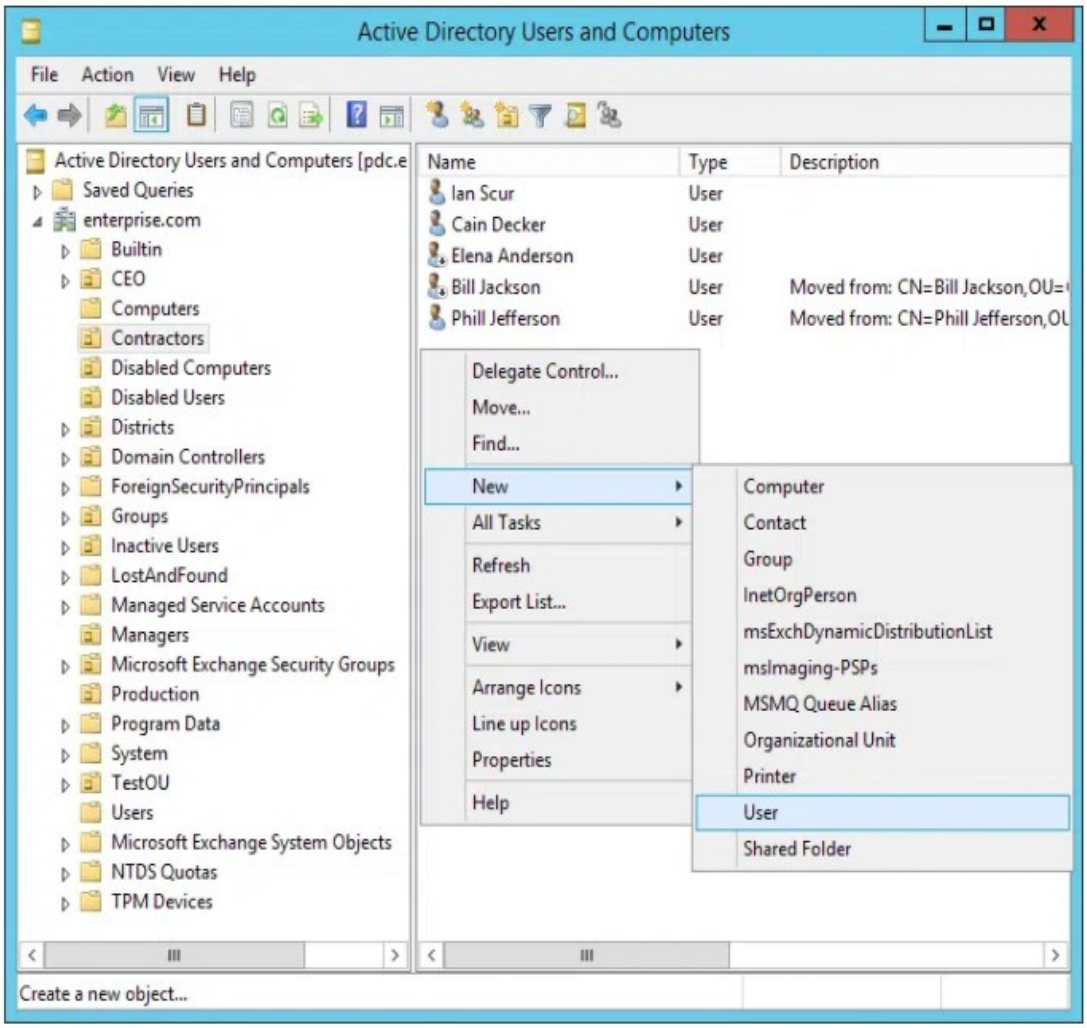
Traditionally ADUC is where most day to day tasks happen. It is here where you can **Pre-stage** the objects in the Organizational Units (OU) (folders) for your computers, which will not only save administrative time later but also make the domain join faster and cleaner because the object is already where you want it to be. Several organizations have a security practice that when a computer is joined to the Domain without being pre-staged it will automatically be moved to Limbo (an area that will cause the computer to not be available until it is moved by a higher echelon administrator to the proper unit's folder. Other tasks include defining user profile information, passwords, and grant access to network resources. ADUC is the most organized and easiest area to perform these tasks but a new program is now available that can do everything that ADUC can do and more

Why use Active Directory Users and Computers (ADUC)?

ADUC is a Microsoft Management Console (MMC) snap-in that enables administrators to manage Active Directory objects and their attributes. For example, they can:

- Change passwords
- Reset user accounts
- Add users to security groups
- Create and delete organizational units (OUs)
- Handle FSMO roles like RID master, PDC Emulator and infrastructure master
- Create and manage computers, groups and users and their attributes
- Delegate control of objects
- Define advanced security and auditing in AD





ADUC Advanced Features

If you enable the Advanced Features setting in ADUC (using the View window), you can also manage:

- LostAndFound container
- NTDS quotas
- Program data
- System information
- Enabling Advanced Features adds many tabs to the properties page of an object, including Published Certificates, Attribute Editor and Password Replication.

Active Directory Groups

The Active Directory groups are a collection of Active Directory objects. The group can include users, computers, other groups, and other AD objects. The administrator manages the group as a single object. In Windows, there are 2 types of groups (**Security and Distribution**): with three scopes in each and a local security group.

Active Directory groups can be used:

• To simplify the administration by assigning share (resource) permissions to a group rather than individual users. When you assign permissions to a group, all of its members have the same access to the resource;
• To delegate the control by assigning user rights to a group using Group Policies. In the future, you can add new members to the group who need the permissions granted by this group;
• To create email distribution lists.

Active Directory Sites and Services

Configuring Active Directory Sites & Services is an important, but sometimes mishandled, part of administering a domain. The information in Sites & Services is used by AD to create a replication topology, determine which DCs should authenticate clients, and help users connect to applications and services. Sites & Services being misconfigured can result in increased logon times, poor user experience with some applications, or worse - a **tombstoned** DC if issues go undetected for too long. Here, we'll review what the information in Sites & Services is used for, the correct way to set it up, and how to verify that it is configured correctly.

A directory service site topology is a logical representation of the physical network. Designing a site topology for Active Directory Domain Services (ADDS) involves planning for domain controller placement and designing sites, subnets, site links, and site link bridges to ensure efficient routing of query and replication traffic. Designing a site topology helps administrators efficiently route client queries and Active Directory replication traffic. A well-designed site topology helps any organization achieve the following benefits:

- Minimize the cost of replicating Active Directory data.
- Minimize administrative efforts that are required to maintain the site topology.
- Schedule replication that enables locations with slow or dial-up network links to replicate Active Directory data during off-peak hours.

- Optimize the ability of client computers to locate the nearest resources, such as domain controllers and Distributed File System (DFS) servers. This helps to reduce network traffic over slow wide area network (WAN) links, improve logon and logoff processes, and speed up file download operations.

Functions of ADSS

Primarily, AD Sites & Services provides a way for the administrator of a domain to map the physical layout of the network in a way that Active Directory can understand. This information is used to help the domain controllers and client machines in the following ways:

DC Replication Topology - Determining which DCs are allowed to replicate with each other and creating a full topology that ensures all DCs can replicate with the rest of the domain.
Client DC Selection - Ensuring that client devices (servers and workstations) choose the nearest DC for authentication and other domain communications
Distributed File Share (DFS) Namespaces - Ensuring that clients requesting access to a DFS Namespace share are accessing the nearest copy of the data when possible
Other Site-Aware Services - Helping client machines communicate with the correct server for applications like Exchange

Typically ADSS does not require manual configuration but it can be done if needed. Now that we know what the information in ADSS is used for, we can understand why it's important to configure it correctly. At its core, the topology the administrator defines in ADSS should match the physical and network topology of their environment. The topology is made up of the following items:

AD Sites - These are the objects that represent the physical locations of DCs and/or client machines. Typically, every defined site should have at least one DC, and the sites should match the physical locations the servers are in. Specifically, servers that are on the same LAN should be in the same site, and servers that are separated by a WAN should be in separate sites.
Subnets - Subnets are the IP ranges that are associated with specific AD sites. DCs and clients use the subnets administrators define to determine what site they are in. For example, if they define a subnet of 192.168.0.0/24 and map it to the US-NYC site, then any server or workstation with an IP address in that network will know it's located in that site.

Site Links - Site links connect 2 or more AD sites together and should match the physical network connections among the sites. For example, if the organization has 3 locations that have full connectivity to each other, then an administrator could have a single site link that contains all 3 sites. This would tell Active Directory and the related services that all locations can and should communicate directly with each other.

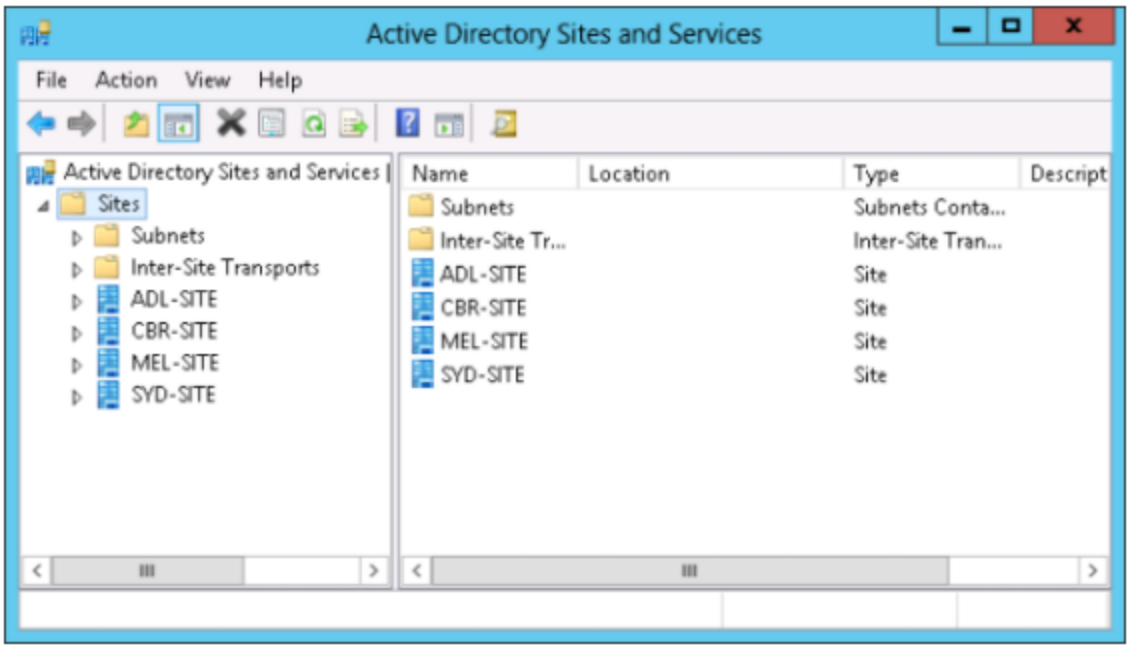
The most common issues are within the site links. It's easy to overlook how the site links should be configured, but it is very important to set them up correctly. The site links help the

KCC (Knowledge Consistency Checker) decide which DCs should be replicating with each other. ***The KCC is an automatic function within AD that has multiple tasks***, one of which is ensuring that all DCs have suitable replication partners. AD site links will typically follow one of two patterns:

- **Full Mesh** - In a full mesh topology, all sites are connected to each other over relatively similar link types. For a full mesh topology, a unit can have a single site link that contains all of the defined AD sites.
- **Hub and Spoke** - A hub and spoke topology has one or more hub sites and multiple branch sites that communicate only with the nearest hub location. Typically, the hubs are central datacenters, while the branch sites would be branch office locations or smaller, remote datacenters.

The most important part of configuring the site links is ensuring that they match the network connectivity of the sites. If administrators create a site link and add sites to it that can't all communicate with each other, they'll likely end up with replication errors. The KCC assumes that all sites in each site link have full connectivity to each other, and if that isn't the case, they can run into issues.

The AD Sites & Services configuration plays a key role in certain Active Directory processes but is often overlooked or not done correctly. In addition to the above points, there are many more components to the configuration, such as site link costs, site link bridging, and the Bridge All Site Links option. Overall, ensuring the site topology is correct and there aren't any KCC or replication errors is a great first step to healthy AD replication.



Domain Name System (DNS)

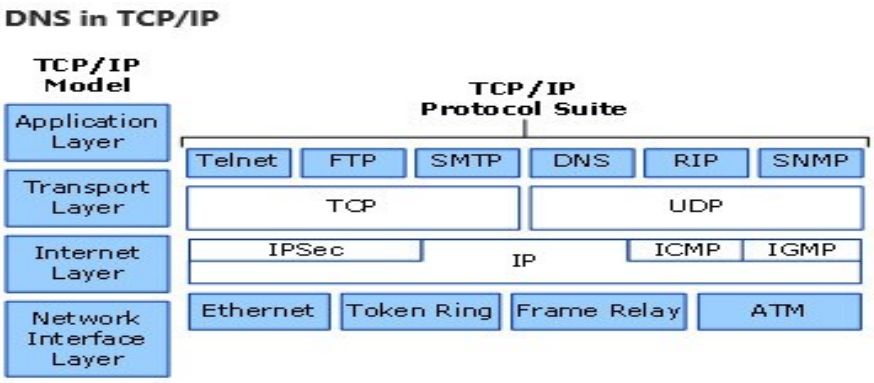
Domain Name System (DNS) is one of the industry-standard suite of protocols that comprise TCP/IP, and together the DNS Client and DNS Server provide computer name-to-IP address mapping name resolution services to computers and users.

In Windows Server, DNS is a **server role** that administrators can install by using Server Manager or Windows PowerShell commands. If they are installing a new Active Directory forest and domain, DNS is automatically installed with Active Directory as the **Global Catalogue server** for the forest and domain.

Active Directory Domain Services (ADDS) uses DNS as its domain controller location mechanism. When any of the principal Active Directory operations is performed, such as authentication, updating, or searching, computers use DNS to locate Active Directory domain controllers. In addition, domain controllers use DNS to locate each other.

The DNS Client service is included in all client and server versions of the Windows operating system, and is running by default upon operating system installation. When an administrator configures a TCP/IP network connection with the IP address of a DNS server, the DNS Client queries the DNS server to discover domain controllers, and to resolve computer names to IP addresses. For example, when a network user with an Active Directory user account logs in to an Active Directory domain, the DNS Client service queries the DNS server to locate a domain controller for the Active Directory domain. When the DNS server responds to the query and provides the domain controller's IP address to the client, the client contacts the domain controller and the authentication process can begin.

The Windows Server DNS Server and DNS Client services use the DNS protocol that is included in the TCP/IP protocol suite. DNS is part of the application layer of the TCP/IP reference model, as shown in the following illustration.



Application Load Balancing- When there are multiple instances of an application at different locations, administrators can use DNS policy to balance the traffic load between the different application instances, dynamically allocating the traffic load for the application.

Split Brain DNS- With split-brain DNS, DNS records are split into different Zone Scopes on the same DNS server, and DNS clients receive a response based on whether the clients are internal or external clients. They can configure split-brain DNS for Active Directory integrated zones or for zones on standalone DNS servers.

Forensics- Administrators can use DNS policy to redirect malicious DNS clients to a non-existent IP address instead of directing them to the computer they are trying to reach. This is called the “DNS black hole”

DNS Records

Static record — is a record that doesn't have timestamp information (it will instead have the word "static" printed on its timestamp properties). Static record can only be updated or deleted from the DNS server on-command by the administrator or another program.

Dynamic record — is a record with timestamp information. Usually created automatically by a client machine that obtains IP address from a DHCP server, or can also be created manually in the DNS server. The timestamp indicates the record validity period, if the record does not refreshed or updated after it passed the time specified in the timestamp then it could be subject for deletion when DNS scavenging event occurs.

Host (A) record- Host record or A record is the most basic type of record that exists in the DNS server. The A-record serves the basic function of DNS server which is to map a hostname string into its IP address. In A-record properties, the "**Name**" field contains information about the hostname and the "**Data**" field will contain the IP address information of the host. For example: "the IP address of host named N01MEUDC01 in zone named 01MEU.USMC.MIL is 10.152.80.2".

A-record can only be created and stored in a forward lookup zone and can be created as a static or dynamic record. Every Host A-record should have a corresponding **Pointer record** in the reverse lookup zone.

AAAA record (Quad A) - AAAA record is very similar to A record, the only difference is that A record is using IPv4 and AAAA record is using IPv6. Every provision that applies in A record applies to AAAA record too, except that AAAA record is mapping a hostname string into its IPv6 address.

Pointer (PTR) record - Pointer or PTR record is the counterpart of an A-record. It is essentially the reverse of A record where the "Name" field contains the IP address number and the "Data" field contains the hostname string. PTR record is used to map an IP address into its associated hostname, usually used to find out what host name is using the specified IP address.
For example: "the associated host name that is using IP address of 10.152.80.2 is N01MEUDC01".

The PTR record can only be stored in a **reverse lookup zone**. It can be created manually or automatically as part of the A-record creation. When PTR record is created as part of A-record creation, it will be either static or dynamic record following the A-record.

Note- If the Reverse lookup zone is not created then the Pointer records will not be automatically created and you will have to manually enter them after the reverse lookup zone is created.

DNS hostname resolution and reverse lookup resolution can be seen in action using the Command Prompt or PowerShell when the "**nsLookup"** command is used. Assuming the records have been entered correctly, type nslookup and the hostname it will return the IP address. Alternately, if administrators type nslookup and the IP address it will return the hostname.

Alias (CNAME) record- Alias or CNAME is the mapping of a hostname into another hostname. CNAME is used to create an alias for an existing A record. For example, an alias "MAIL.01MEU.USMC.MIL" can be used to point to an existing A-record for the mail server "N01MEUEX01.01MEU.USMC.MIL" where the service is hosted. This will help to keep the actual mail server name more private.

It is possible to create CNAME record in both forward and reverse lookup zone, either as static or dynamic record. Multiple CNAME can be created pointing to the same A record, but the same CNAME cannot be created more than once. The other thing worth noting is that CNAME can be created in a different zone name than the A record.

Mail Exchanger (MX) record- MX record is used specifically for the purpose of sending email. MX record existence indicates that the domain is willing to receive email. For example, when a mail is about to be sent to recipient name aaron.smith@01MEU.USMC.MIL, the sending server will then try to find out if any MX record exists in the parent domain 01MEU.USMC.MIL.

Administrators can create MX record only in forward lookup zone, either as static or dynamic record, and should point to a valid A record or CNAME of the host where the email service is running. For example, an MX record "01MEU.USMC.MIL" is found on the domain 01MEU.USMC.MIL, and it points to a CNAME record MAIL.01MEU.USMC.MIL which ultimately points to an A record N01MEUEX01.01MEU.USMC.MIL. Multiple MX records using the same "Name" can be created, and each record may point to different A record. Different priority number can be given to each record, and lower number means more prioritized. This is especially helpful when using **DNS round robin** to do load balancing between multiple exchange servers.

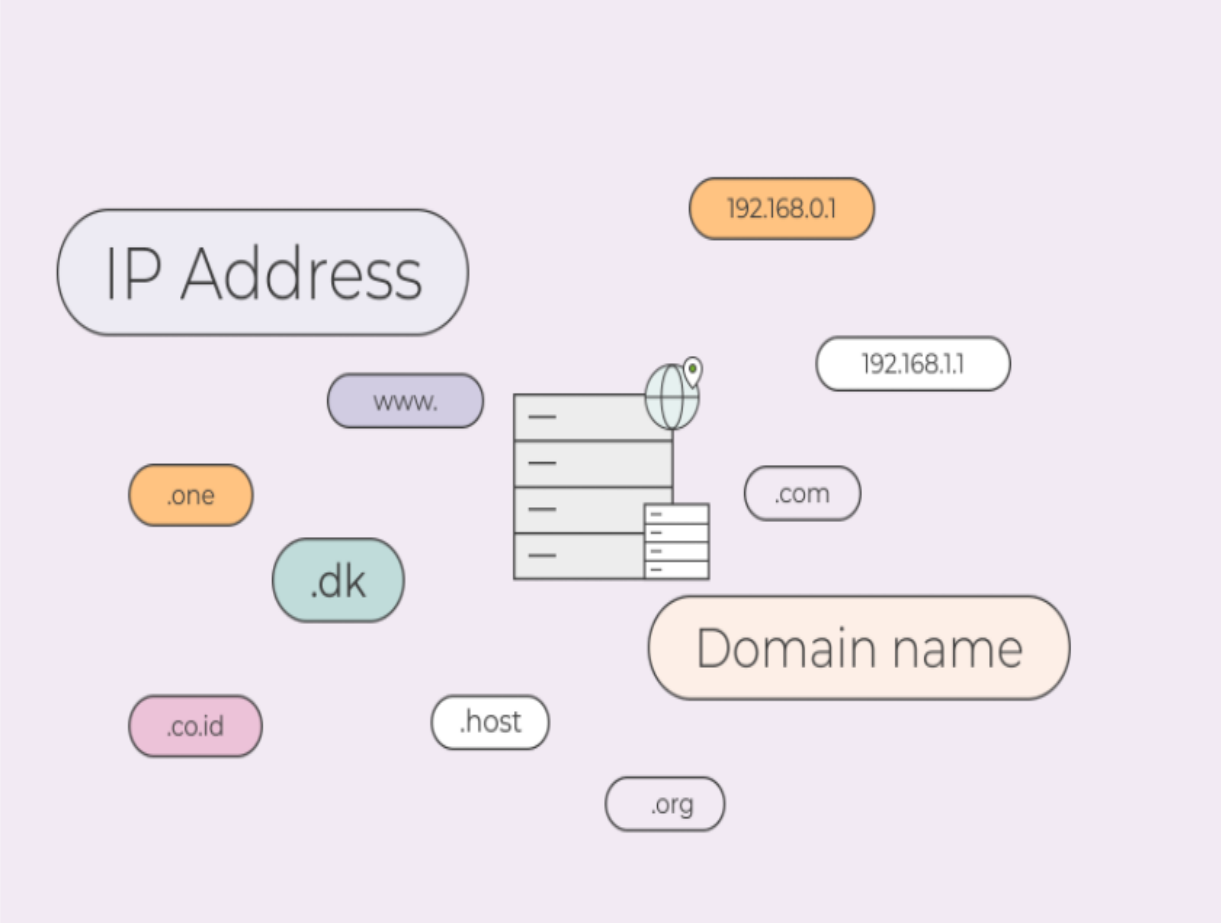
Nameserver (NS) record- NS or Nameserver record is another special type of DNS record. The NS record is used to point to the name of the server where a zone or domain name record is registered. The NS record is created automatically when administrators create or add name servers to their domain.

Start of Authority (SOA) record- SOA or Start of Authority, is a DNS record that contains the name of the server that holds all the information regarding a record. It includes authoritative information such as server identification number, the responsible mail address, the primary name server, etc. The server referenced in the SOA record is assumed to be in charge of any process on the related record. The SOA record is created automatically when administrators create or add name servers to their domain.

Every zone or domain must have an SOA record. When administrators create a new zone or domain in a DNS server, SOA record is created automatically.

Service (SRV) record- SRV or Service record, is a type of DNS record that is used for service discovery. SRV record provides information of what services are available in a particular zone or domain. SRV record has a format for its “Name” value where it starts by the name of the service, name of the protocol, then name of the zone or domain. The “Data” value will contains several information such as the priority, weight, service port number, and the host name where the service is residing. *Cisco Jabber chat solution is heavily dependent on SRV records for DNS resolution.

Text (TXT) record- TXT or Text record is a special type of DNS record. It is not used to forward traffic but rather used to provide information about the domain. TXT can be created in any zone as a static or dynamic record. The “Name” value and the “Data” value can be filled with anything as required by a program or protocol.



Domain Controller Roles

A Domain Controller (DC) is a server that is running a version of the Windows Server operating system and has the Active Directory Domain Services (ADDS) role installed.

When administrators install Windows Server on a computer, they can choose to configure a specific server role for that computer. When administrators want to create a new forest, a new domain, or an additional domain controller in an existing domain, they configure the server with the role of domain controller by installing Active Directory Domain Services (ADDS).

By default, a domain controller stores one domain directory partition consisting of information about the domain in which it is located, plus the schema and configuration directory partitions for the entire forest. There are specialized domain controller roles that perform specific functions in an ADDS environment. These specialized roles include global catalog servers and operations masters.

- **Global Catalog Servers**

Every domain controller stores the objects for the domain in which it is installed. However, a domain controller designated as a global catalog server stores the objects from all domains in the forest. For each object that is not in the domain for which the global catalog server is authoritative as a domain controller, a limited set of attributes is stored in a partial replica of the domain. Therefore, a global catalog server stores its own full, writable domain replica (all objects and all attributes) plus a partial, read-only replica of every other domain in the forest. The global catalog is built and updated automatically by the ADDS replication system. The object attributes that are replicated to global catalog servers are the attributes that are most likely to be used to search for the object in ADDS. The attributes that are replicated to the global catalog are identified in the schema as the partial attribute set (PAS) and are defined by default by Microsoft.

The global catalog makes it possible for clients to search ADDS without having to be referred from server to server until a domain controller that has the domain directory partition storing the requested object is found. By default, ADDS searches are directed to global catalog servers. The first domain controller in a forest is automatically created as a global catalog server.

- **Flexible Single Master Operations Master (FSMO) roles**

Domain controllers that hold operations master roles are designated to perform specific tasks to ensure consistency and to eliminate the potential for conflicting entries in the Active Directory database. ADDS defines five operations master roles: the **schema master**, **domain naming master**, **relative identifier (RID) master**, **primary domain controller (PDC) emulator**, and **infrastructure master**.

The following operations masters perform operations that must occur **on only one domain controller in the forest:**

• Schema master

• Domain naming master

The following operations masters perform operations that must occur **on only one domain controller in a domain:**

• Primary Domain Controller (PDC) emulator

• Infrastructure master

• Relative ID (RID) master

There is only **one Schema master per forest**. It is responsible for processing updates to AD schema. This DC is the only one that can process updates to the directory schema. Once the Schema update is complete, it is replicated from the schema master to all other DCs in the directory. It is located by using the AD Schema snap-in of the MMC. *Note that the Schema snap-in is only available if you register the schmmgmt.dll using PowerShell. The likelihood of a entry level 0671 administering the Schema master is slim to none, this is typically done by a NCO, SNCO or even Chief Warrant Officer.

There is only **one Domain naming master per forest**. It is responsible for managing the addition and removal of domains and application partitions in the forest. It handles references to domains in forests that have a trust relationship with the source forest. This DC is the only one that can add or remove a domain from the directory. It can also add or remove cross references to domains in external directories.

There is **one PDC emulator per domain**. It is responsible for replicating password changes. It also performs time synchronization across domain members and is responsible for managing GPOs. The purpose of the time service is to ensure that the Windows Time service uses a hierarchical relationship that controls authority and does not permit loops to ensure appropriate common time usage. It can also provide clients with a list of workgroups and domains. The PDC emulator role holder also retains the following functions: Password changes, Account lockouts, and Authentication failures.

There is **one Infrastructure master (IM) per domain**. It keeps track of changes made in other domains in the forest and their impact on objects in the local domain. In large organizations, it is not recommended to be placed on a domain controller running the global catalog service.

The domain controller assigned the infrastructure master role is responsible for updating the group-to-user references whenever the members of groups are renamed or changed. At any time, there can be only one domain controller acting as the infrastructure master in each domain.

When administrators rename or move a member of a group (and the member resides in a different domain from the group), the group might temporarily appear not to contain that member. The infrastructure master of the group's domain is responsible for updating the group so it knows the new name or location of the member. The infrastructure master distributes the update via multi-master replication.

- The **IM** is responsible for updating cross-domain object references each DC in the Domain, to do that it needs to check for changes on an available GC, then compares its information with the information that the GC has, if any changes, then updates its local information, and updates cross-domain object references each DC in the Domain.

There is **one RID master per domain**. The RID master FSMO role holder is the single DC responsible for processing RID Pool requests from all DCs within a given domain. It is also responsible for removing an object from its domain and putting it in another domain during an object move.

When a DC creates a security principal object such as a user or group, it attaches a unique Security ID (SID) to the object. This SID consists of a domain SID (the same for all SIDs created in a domain), and a relative ID (RID) that is unique for each security principal (object) SID created in a domain.

Each Windows DC in a domain is allocated a pool of RIDs that it is allowed to assign to the security principals it creates. When a DC's allocated RID pool falls below a threshold, that DC issues a request for additional RIDs to the domain's RID master. The domain RID master responds to the request by retrieving RIDs from the domain's unallocated RID pool and assigns them to the pool of the requesting DC. **There is one RID master per domain in a directory.**

FSMO BEST PRACTICES.

Rule 1: The PDC Emulator and RID Master roles should be on the same machine because the PDC Emulator is a large consumer of RIDs.

Tip: Since the PDC Emulator is the role that does the most work by far of any FSMO role, if the machine holding the PDC Emulator role is heavily utilized then move this role and the RID Master role to a different DC, preferable not a global catalog server (GC) since those are often heavily used also.

Rule 2: The Infrastructure Master should **NOT** be placed on a GC.

Tip: Make sure the Infrastructure Master has a GC in the same site as a direct replication partner.

-**Exception 1:** It's OK to put the Infrastructure Master on a GC if your forest has only one domain.

-**Exception 2**: It's OK to put the Infrastructure Master on a GC if every DC in your forest has the GC.

Rule 3: For simpler management, the Schema Master and Domain Naming Master can be on the same machine, which should also be a GC.

Rule 4: Proactively check from time to time to confirm that all FSMO roles are available or write a script to do this automatically.

Tip: If any FSMO role holders at a remote site are unavailable, check first to see if the WAN link is down.

Transferring FSMO Roles

Transferring the RID master, PDC emulator and Infrastructure Master Roles.

1. Log in to the windows server as domain administrator.
2. Click on Server Manager > Tools > Active Directory Users and Computers.
3. In MMC, right click on the domain name then click on Operation Masters.
4. In next window it will show the 3 FSMO roles. The default is RID. In there it shows the current RID holder. Then it is asking if you need to change it to a new windows server. .
5. Click on Change.
6. Select either RID, PDC, or Infrastructure. If you need to transfer the role to another DC, select Change… and then search for the new DC you want the role transferred to.
7. Select OK.

Transfer Schema Master Role

1. Log in to the windows server as domain administrator.
2. Open the Run window in server (Windows key + R) and type regsvr32 schmmgmt.dll and press enter.
3. Open new Microsoft Management Console (mmc) and then from the snap in, select Active Directory Schema and click on the Add button then OK.

4. Once you have the schema, right click Active Directory Schema and click Change Active Directory Domain Controller to select your DC.
5. Then again, Right click Active Directory Schema > Operation Master.
6. In here it shows the current schema master role holder and it's asking if we need to move it to another windows server. Click Change to move the role over.

Transfer or Seizing FSMO Roles

There may come a time when a Domain Controller holding a role becomes unrecoverable. In this instance you are able to seize that role using another Domain Controller through PowerShell.

The FSMO roles are moved to another Domain Controller by using the command:

```
Move-ADDirectoryServerOperationMasterRole
```

- **Transferring all roles**

```
Move-ADDirectoryServerOperationMasterRole -Identity "Target-DC"
-OperationMasterRole SchemaMaster,RIDMaster,InfrastructureMaster
,DomainNamingMaster,PDCEmulator
```

- **Seizing all roles:**

```
Move-ADDirectoryServerOperationMasterRole -Identity "Target-DC"
-OperationMasterRole SchemaMaster,RIDMaster,InfrastructureMaster
,DomainNamingMaster,PDCEmulator -Force
```

For example, the target Domain Controller name is IMEFDC01.
We use this command to transfer all roles to another Domain Controller:

```
Move-ADDirectoryServerOperationMasterRole -Identity "IMEFDC01" -
OperationMasterRole SchemaMaster,RIDMaster,InfrastructureMaster,
DomainNamingMaster,PDCEmulator
```

We use this command to seize all roles only from a permanently offline Domain Controller:

```
Move-ADDirectoryServerOperationMasterRole -Identity "IMEFDC01" -
OperationMasterRoleSchemaMaster,RIDMaster,InfrastructureMaster,
DomainNamingMaster,PDCEmulator -Force
```

Instead of typing the Names of the operations master roles, Numbers may also be specified.

Role Name	Role Number
PDCEmulator	0
RIDMaster	1
InfrastructureMaster	2
SchemaMaster	3
DomainNamingMaster	4

Transferring all roles:

```
Move-ADDirectoryServerOperationMasterRole -Identity "Target-DC"
-OperationMasterRole 0,1,2,3,4
```

Seizing all roles:

```
Move-ADDirectoryServerOperationMasterRole -Identity "Target-DC"
-OperationMasterRole 0,1,2,3,4 -Force
```

For example, my target Domain Controller name is **IMEFDC01**.

We will use this command to transfer roles to another Domain Controller:

```
Move-ADDirectoryServerOperationMasterRole -Identity "IMEFDC01" -
OperationMasterRole 0,1,2,3,4
```

We use this command to seize roles to another Domain Controller:

```
Move-ADDirectoryServerOperationMasterRole -Identity "IMEFDC01" -
OperationMasterRole 0,1,2,3,4 –Force
```

We use this command to transfer only one selected role to another Domain Controller:

```
Move-ADDirectoryServerOperationMasterRole -Identity "IMEFDC01" -
OperationMasterRole SchemaMaster
```

or

```
Move-ADDirectoryServerOperationMasterRole -Identity "IMEFDC01" -
OperationMasterRole 3
```

Active Directory Federation Services (ADFS)

What is ADFS?

Active Directory Federation Services (ADFS) is a Single Sign-On (SSO) solution created by Microsoft. As a component of Windows Server operating systems, it provides users with authenticated access to applications that are not capable of using Integrated Windows Authentication (IWA) through Active Directory (AD).

Developed to provide flexibility, ADFS gives organizations the ability to control their employees' accounts while simplifying the user experience: employees only need to remember a single set of credentials to access multiple applications through SSO.

How does ADFS work?

ADFS manages authentication through a proxy service hosted between AD and the target application. It uses a Federated Trust, linking ADFS and the target application to grant access to users. This enables users to log onto the federated application through SSO without needing to authenticate their identity on application directly.

Why do companies use ADFS?

ADFS was born out of the need to overcome the authentication challenges created by AD in an increasingly connected online world. AD and IWA have set limitations when it comes to modern authentication, and cannot authenticate users accessing AD integrated applications externally.

This is a challenge in the modern workplace, where users often need to access applications that are not owned or managed by their AD organization.
ADFS is able to resolve and simplify these third-party authentication challenges, but does come with certain risks and disadvantages.

ADFS solves the problem of users who need to access AD integrated applications while working remotely, offering a flexible solution whereby they can authenticate using their standard organizational AD credentials via a web interface. It allows users from one organization to access the applications of another organization beyond the realm of their AD domain. Examples include applications in a partner organization or modern cloud services, which now form part of many organizations' extended IT landscape.
Over 90% of organizations use Active Directory, which means many use ADFS as well.
Good examples of applications that are ADFS integrated are; Cisco Jabber, Solarwinds, Web Help Desk, SharePoint, and even Transverse Chat.

Dynamic Host Configuration Protocol (DHCP)

Dynamic Host Configuration Protocol (DHCP) is a client/server protocol that automatically provides an Internet Protocol (IP) host with its IP address and other related configuration information such as the subnet mask and default gateway. RFCs 2131 and 2132 define DHCP as an Internet Engineering Task Force (IETF) standard based on Bootstrap Protocol (BOOTP), a protocol with which DHCP shares many implementation details. DHCP allows hosts to obtain required TCP/IP configuration information from a DHCP server.

Windows Server 2019 includes DHCP Server, which is an optional networking server role that administrators can deploy on their network to lease IP addresses and other information to DHCP clients. All Windows-based client operating systems include the DHCP client as part of TCP/IP, and DHCP client is enabled by default.

Why use DHCP?

Every device on a TCP/IP-based network must have a unique unicast IP address to access the network and its resources. Without DHCP, IP addresses for new computers or computers that are moved from one subnet to another must be configured manually; IP addresses for computers that are removed from the network must be manually reclaimed.

With DHCP, this entire process is automated and managed centrally. The DHCP server maintains a pool of IP addresses and leases an address to any DHCP-enabled client when it starts up on the network. Because the IP addresses are dynamic (leased) rather than static (permanently assigned), addresses no longer in use are automatically returned to the pool for reallocation.

The network administrator establishes DHCP servers that maintain TCP/IP configuration information and provide address configuration to DHCP-enabled clients in the form of a lease offer. The DHCP server stores the configuration information in a database that includes:

- Valid TCP/IP configuration parameters for all clients on the network.
- Valid IP addresses, maintained in a pool for assignment to clients, as well as excluded addresses.
- Reserved IP addresses associated with particular DHCP clients. This allows consistent assignment of a single IP address to a single DHCP client.
- The lease duration, or the length of time for which the IP address can be used before a lease renewal is required.

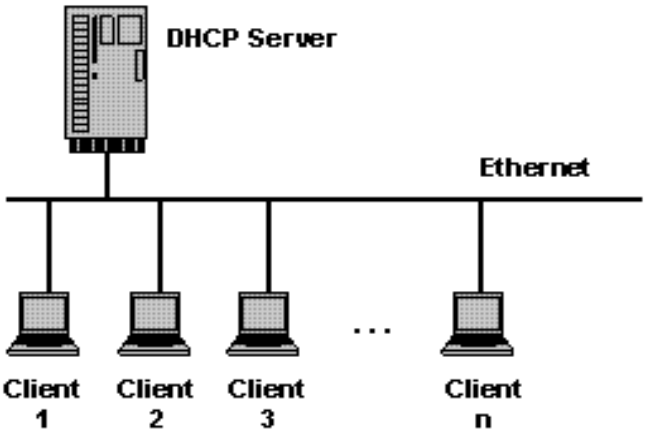
A DHCP-enabled client, upon accepting a lease offer, receives:

- A valid IP address for the subnet to which it is connecting.
- Requested DHCP options, which are additional parameters that a DHCP server is configured to assign to clients. Some examples of DHCP options are Router (default gateway), DNS Servers, and DNS Domain Name.

Benefits of DHCP

DHCP provides the following benefits.

- **Reliable IP address configuration**. DHCP minimizes configuration errors caused by manual IP address configuration, such as typographical errors, or address conflicts caused by the assignment of an IP address to more than one computer at the same time.
- **Reduced network administration**. DHCP includes the following features to reduce network administration:
  - Centralized and automated TCP/IP configuration.
  - The ability to define TCP/IP configurations from a central location.
  - The ability to assign a full range of additional TCP/IP configuration values by means of DHCP options.
  - The efficient handling of IP address changes for clients that must be updated frequently, such as those for portable devices that move to different locations on a wireless network.
  - The forwarding of initial DHCP messages by using a DHCP relay agent, which eliminates the need for a DHCP server on every subnet.



DHCP Server assigns IP addresses to multiple clients

Print and Document Services

Print and Document Services enables the ability to centralize print server and network printer tasks. With this role, administrators can also receive scanned documents from network scanners and route the documents to a shared network resource, Windows SharePoint Services site, or email addresses.

IP Address Management (IPAM)

IP Address Management (IPAM) is an integrated suite of tools to enable end-to-end planning, deploying, managing and monitoring of the IP address infrastructure, with a rich user experience. IPAM automatically discovers IP address infrastructure servers and Domain Name System (DNS) servers on the network and enables administrators to manage them from a central interface.

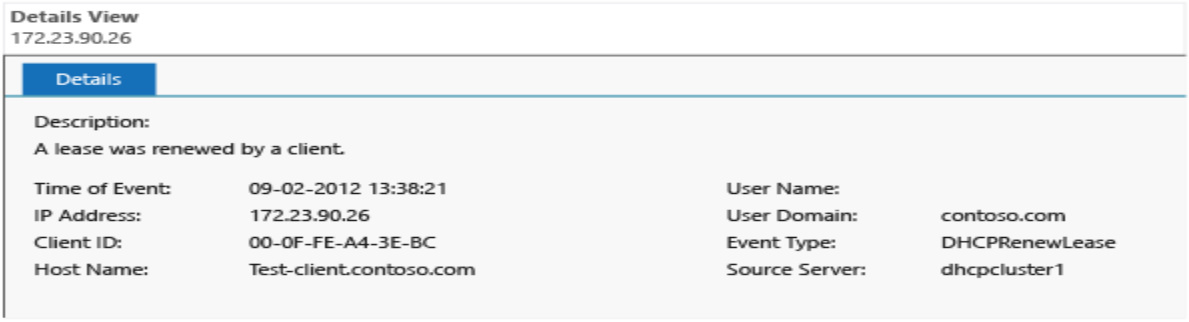
Why use IPAM?

Network Audit

IPAM's audit feature provides a centralized repository for all configuration changes performed on DHCP servers and the IPAM server, and for IP addresses issued on the network. IPAM audit tools enable administrators to view potential configuration problems on DHCP servers by actively tracking and reporting all administrative actions. Detailed IP address tracking data is also provided, including client IP addresses, client ID, host name, and user name. Advanced search capabilities enables the ability to selectively search for events and obtain results that associate user logons to specific devices and times.

Key features of network audit include the following.

- Query the event catalog for DHCP configuration changes across multiple servers from a single console
- Track users, devices, and IP addresses for specified intervals with advanced queries using DHCP lease logs and logon events from domain controllers and network policy servers
- Track and report changes made to the IPAM server
- Export audit findings and create reports
- Quickly resolve configuration problems and track service level agreements

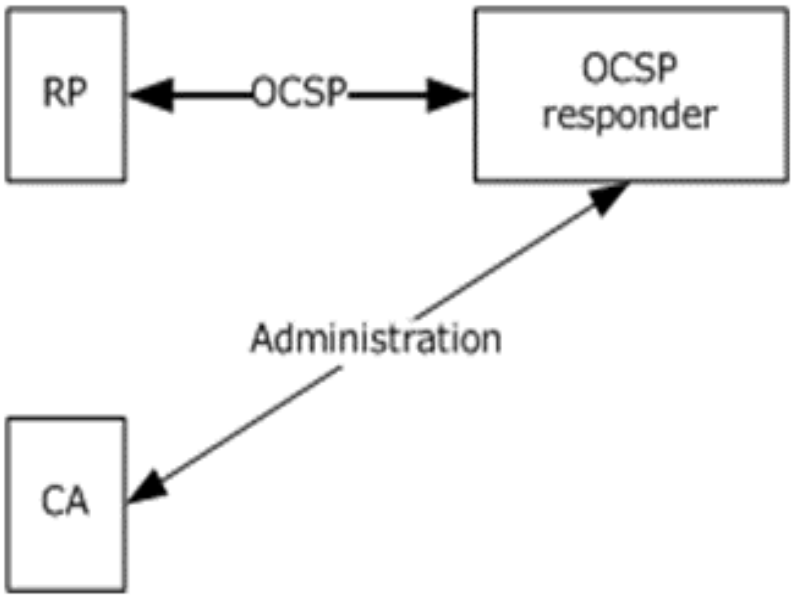


Microsoft Online Certificate Status Protocol (OCSP)

Online Certificate Status Protocol (OCSP) provides an efficient mechanism for distributing certificate revocation information. When certificates are exchanged and validated, computers need to determine if the certificate has been revoked – meaning the CA has reason to consider the certificate as untrusted. This often placed in a Certificate Revocation List (CRL). Clients download this potentially large CRL just to find a small serial number of a certificate to determine if it has been revoked. OCSP on the other hand changes the process to a SQL like process where clients send a secure query to an OCSP Responder (server) and ask if the serial number it is looking at has been marked as revoked.

OCSP is a component of a public key infrastructure (PKI). A PKI consists of a system of digital certificates, certification authorities (CAs), and other registration authorities (RAs) that verify and authenticate the validity of each party involved in an electronic transaction through the use of public key cryptography.

The certificate status received as a result of using OCSP is known as a response from an OCSP responder. The OCSP request/response process involves a number of different machines (or functions that might be hosted on the same machine), as indicated in Figure 1



File and Storage Services

File and Storage Services includes technologies that help you set up and manage one or more file servers, which are servers that provide central locations on your network where you can store files and share them with users. If your users need access to the same files and applications, or if centralized backup and file management are important to your organization, you should set up one or more servers as a file server by installing the File and Storage Services role and the appropriate role services.

The File and Storage Services role and the Storage Services role service are installed by default, but without any additional role services. This basic functionality enables you to use Server Manager or Windows PowerShell to manage the storage functionality of your servers. However, to set up or manage a **file server**, you should use the **Add Roles and Features Wizard** in Server Manager or the **Install-WindowsFeature PowerShell cmdlet** to install additional File and Storage Services role services.

Practical applications

Administrators can use the **File and Storage Services role** to set up and manage multiple file servers and their storage capabilities by using Server Manager or Windows PowerShell. Some of the specific applications include the following:

- **Storage Spaces** - Use to deploy high availability storage that is resilient and scalable by using cost-effective industry-standard disks.

- **Folder Redirection, Offline Files, and Roaming User Profiles** - Use to redirect the path of local folders (such as the Documents folder) or an entire user profile to a network location, while caching the contents locally for increased speed and availability.

- **Work Folders** - Use to enable users to store and access work files on personal PCs and devices, in addition to corporate PCs. Users gain a convenient location to store work files and access them from anywhere. Organizations maintain control over corporate data by storing the files on centrally managed file servers and optionally specifying user device policies (such as encryption and lock screen passwords).

- **Data Deduplication** - Use to reduce the disk space requirements of your files, saving money on storage.

- **iSCSI Target Server** - Use to create centralized, software-based, and hardware-independent iSCSI disk subsystems in storage area networks (SANs).
- **Server Manager** - Use to remotely manage multiple file servers from a single window.
- **Windows PowerShell** - Use to automate the management of the majority of administration tasks for file servers.

Active Directory Administration

Remote Server Administrative Tools (RSAT)

Remote Server Administrative Tools (RSAT) lets IT admins manage Windows Server roles and features from a Windows Workstation. Remote Server Administration Tools includes Server Manager, Microsoft Management Console (mmc) snap-ins, consoles, Windows PowerShell cmdlets and providers, and some command-line tools for managing roles and features that run on Windows Server.

Remote Server Administration Tools includes Windows PowerShell cmdlet modules that can be used to manage roles and features that are running on Remote servers. Although Windows PowerShell remote management is enabled by default on Windows Server, it is not enabled by default on Windows 10. To run cmdlets that are part of Remote Server Administration Tools against a Remote server, run `Enable-PSremoting` in a Windows PowerShell session that has been opened with elevated user rights (that is, **Run as Administrator**) on your Windows client computer after installing Remote Server Administration Tools.

Active Directory Administration Center (ADAC)

ADUC is the most organized and easiest area to perform these administrative tasks but a new program is now available that can do everything that ADUC can do but with more capabilities and the ability to have more control over individuals and groups. There is even a feature to be able to recover objects that were accidentally deleted.
ADAC is the newer, primary tool for performing day to day tasks in the administration of the Active Directory environment. ADAC Applies to Windows Server 2008 R2 and newer. ADAC supersedes Active Directory Users and Computers (ADUC) in functionality. Many administrators still prefer to use ADUC for most tasks because the hierarchal structure is easier to navigate.

ADAC User interface features include the:
- Breadcrumb bar – displays the location of the currently selected object
- Navigation pane – enables you to browse Active Directory
- Management lists – Displays the contents of the container that is currently selected
- Preview pane – Displays various information about the object or container that is currently selected
- Tasks pane – enables you to perform different tasks on the object or container that is currently selected in management lists
- Active Directory Recycle Bin - recover deleted objects with the Recycle Bin. (not activated by default.
- Ability to create a Fine Grained Password Policy from ADAC, but you can't do any of this from ADUC.

The Global Search feature in ADAC is used to search for objects in AD to include all domains and not just the global catalog lookup.

ADAC Advanced Features

Active Directory Administration Center has some additional advanced features useful for Systems Administrators such as the AD Recycle Bin and the Fine grained password policy editor.
- AD recycle Bin – provides administrators a way to recover directory objects that were accidentally deleted
- Fine grained password policy – provides administrators a way to set different password policies to groups or users.

To enable the Active Directory Recycle Bin, open the Active Directory Administrative Center and click the name of your forest in the navigation pane. From the Tasks pane, click Enable Recycle Bin. The Deleted Objects container shows you all the restorable objects in that domain partition. It is important to note that by default AD Recycle Bin is disabled.

ADAC in Windows Server makes Fine Grained Password Policy management much easier and simpler due to the graphical user interface. Using ADAC, administrators can create a separate password policy for user accounts with different password needs, much more easily and quickly. First, ensure that your domain is at Window Server 2008 domain functional level or higher. Next, select New Password Settings, under the Password Settings Container which resides under System. Complete the password settings dialog box as desired and then click Add to apply this policy directly to the desired users, groups etc.

Active Directory Role Based Access control (RBAC)

RBAC is a method of regulating access to computer or network resources based on the roles of individual users within an enterprise. In this context, access is the ability of an individual user to perform a specific task, such as view, create, or modify a file. Roles are defined according to job competency, authority, and responsibility within the enterprise.

The principle of least privilege (PoLP; also known as the principle of least authority) is an important concept to use when creating account permissions. It promotes minimal user profile privileges on computers, based on users' job necessities. The principle of least privilege translates to giving people the lowest level of user rights that they can have and still do their jobs.

Example Hierarchy: In this example, Subordinate Units should only have rights to their respective OU. This should be Broken-down by the RCT Admins.

Metadata cleanup

Note- During the labs in the class room you can play with removing domain controllers and the associated metadata clean up but in the CAPSTONE event the instructor in charge of your room will be the one to remove the metadata. If a domain controller does not install properly or you have to delete the Domain controllers VM notify the Instructor to do the cleanup.

Metadata cleanup is a required procedure after a forced removal of a domain controller. You perform metadata cleanup on a domain controller, in the domain of the domain controller that you forcibly removed. Metadata cleanup removes data from ADDS that identifies a domain controller to the replication system. Metadata cleanup also removes File Replication Service (FRS) and Distributed File System (DFS) Replication connections and attempts to transfer or seize any operations master (also known as flexible single master operations or FSMO) roles that the retired domain controller holds. Because you are new to IT you should stick with the GUI methods until you become more familiar with these processes.

There are three options to clean up server metadata:

- Clean up server metadata by using GUI tools
- Clean up server metadata using the command line
- Clean up server metadata by using a script

Note- if you receive an "Access is denied" error when you use any of these methods to perform metadata cleanup, make sure that the computer object and the NTDS Settings object for the domain controller are not protected against accidental deletion. To verify this right-click the computer object or the NTDS Settings object, click Properties, click Object, and clear the Protect object from accidental deletion check box. In Active Directory Users and Computers, the Object tab of an object appears if you click View and then click Advanced Features.

Clean up server metadata using GUI tools

When you use Remote Server Administration Tools (RSAT) or the Active Directory Users and Computers console (Dsa.msc) that is included with Windows Server to delete a domain controller computer account from the Domain Controllers organizational unit (OU), the cleanup of server metadata is performed automatically. Before Windows Server 2008, you had to perform a separate metadata cleanup procedure.

You can also use the Active Directory Sites and Services console (Dssite.msc) to delete a domain controller's computer account, which also completes metadata cleanup automatically. However, Active Directory Sites and Services removes the metadata automatically only when you first delete the NTDS Settings object below the computer account in Dssite.msc.

As long as you are using the Windows Server 2008 or newer RSAT versions of Dsa.msc or Dssite.msc, you can clean up metadata automatically for domain controllers running earlier versions of Windows operating systems.

Membership in Domain Admins, or equivalent, is the minimum required to complete these procedures.

Clean up server metadata using Active Directory Sites and Services

Open Active Directory Sites and Services.
If you have identified replication partners in preparation for this procedure and if you are not connected to a replication partner of the removed domain controller whose metadata you are cleaning up, right-click Active Directory Sites and Services, and then click Change Domain Controller. Click the name of the domain controller from which you want to remove the metadata, and then click OK.
Expand the site of the domain controller that was forcibly removed, expand Servers, expand the name of the domain controller, right-click the NTDS Settings object, and then click Delete.
In the Active Directory Sites and Services dialog box, click Yes to confirm the NTDS Settings deletion.

In the Deleting Domain Controller dialog box, select This Domain Controller is permanently offline and can no longer be demoted using the Active Directory Domain Services Installation Wizard (DCPROMO), and then click Delete.
If the domain controller is a global catalog server, in the Delete Domain Controller dialog box, click Yes to continue with the deletion.
If the domain controller currently holds one or more operations master roles, click OK to move the role or roles to the domain controller that is shown.
Right-click the domain controller that was forcibly removed, and then click Delete.
In the Active Directory Domain Services dialog box, click Yes to confirm the domain controller deletion.

Clean up server metadata using the command line
As an alternative, you can clean up metadata by using Ntdsutil.exe, a command-line tool that is installed automatically on all domain controllers and servers that have Active Directory Lightweight Directory Services (AD LDS) installed. Ntdsutil.exe is also available on computers that have RSAT installed.

To clean up server metadata by using Ntdsutil

******NOTE- NTDSUTIL should be used with extreme caution, improper use can cause irreparable harm to the domain controller if you do not know exactly what you are doing.******
Open a command prompt as an administrator: On the Start menu, right-click Command Prompt, and then click Run as administrator. If the User Account Control dialog box appears, provide credentials of an Enterprise Administrator if required, and then click Continue.

At the command prompt, type the following command, and then press ENTER:

```
ntdsutil
```

At the ntdsutil: prompt, type the following command, and then press ENTER:

```
metadata cleanup
```

At the metadata cleanup: prompt, type the following command, and then press ENTER:

```
remove selected server <ServerName>
```

In Server Remove Configuration Dialog, review the information and warning, and then click Yes to remove the server object and metadata.

At this point, Ntdsutil confirms that the domain controller was removed successfully. If you receive an error message that indicates that the object cannot be found, the domain controller might have been removed earlier.

At the metadata cleanup: and ntdsutil: prompts, type quit, and then press ENTER.

To confirm removal of the domain controller:

Open Active Directory Users and Computers. In the domain of the removed domain controller, click Domain Controllers. In the details pane, an object for the domain controller that you removed should not appear.

Open Active Directory Sites and Services. Navigate to the Servers container and confirm that the server object for the domain controller that you removed does not contain an NTDS Settings object. If no child objects appear below the server object, you can delete the server object. If a child object appears, do not delete the server object because another application is using the object.

REPLICATION

REPADMIN TOOL

Most of the time Active Directory replication seems to work pretty well. When the replication process does break down however, troubleshooting the problem can sometimes be surprisingly difficult. Thankfully, the Windows operating system includes a command line tool called Repadmin that administrators can use to diagnose (and in some cases repair) Active Directory replication.

To use the Repadmin tool, administrators must open an elevated Command Prompt window or PowerShell.

The first command they should try will give them the replication summary. They can also abbreviate the command.

```
Repadmin /replsummary  OR  Repadmin /replsum
```

This command provides administrators with a summary view of the current replication health. As they look at the output, they will notice that it is divided into two main sections -- Source DSA and Destination DSA. Administrators will also notice that the same servers are listed in both sections. The reason for this is that the Active Directory uses a multi master domain model. In other words, Active Directory updates can be written to any domain controller (with read-only domain controllers being the notable exception). Those updates are then replicated to the other domain controllers in the domain. This is the reason why administrators see the same domain controllers listed as both source and destination DSAs. If my domain contained any read only domain controllers, they would only be listed in the Destination DSA section.

Of course the replication summary report does more than just list your domain controllers. It also lists the largest replication deltas. You can also see the total number of replications that have recently been attempted, as well as how many of those attempts failed. You can even see the percentage of attempts that resulted in an error.

The Repadmin /queue command shows you how many items are in the queue waiting to be replicated. It is OK if there is nothing or a few items in the Queue.

```
Repadmin /Queue
```

The Repadmin /Showrepl command displays the GUID of each object that was initially replicated, as well as the replication result. This is useful because you may find that there is a single object for which replication is failing.

```
Repadmin /Showrepl
```

If you suspect that there may be a problem with Active Directory replication the Repadmin command supports about a dozen different switches that you can use to correct a wide variety of problems.

The Repadmin /replicate command is a good place to start when things are not replicating. Repadmin /replicate will manually initiate the replication process.

```
Repadmin /replicate
```